

B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

October 2022 Semester End Main Examinations

Programme: B.E

Branch: Computer Science and Engineering

Course Code: 19CS4PCADA

Course: Analysis and Design of Algorithms

Semester: IV

Duration: 3 hrs.

Max Marks: 100

Date: 14.10.2022

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may be suitably assumed.

UNIT - I

- 1 a) Define Algorithm and explain its characteristics. 06
- b) Build a recursive algorithm to solve Tower of Hanoi problem. Show that the efficiency of this algorithm is exponential. 08
- c) Write an algorithm to find uniqueness of elements in an array and give the mathematical analysis of this non-recursive algorithm with steps. 06

UNIT - II

- 2 a) Illustrate the tracing of Bubble sort algorithm for the following set of numbers and derive its time efficiency 25, 10, 72, 18, 40, 11, 64, 58, 32, 9 07
- b) Differentiate between Depth-First Search (DFS) and Breadth-First Search (BFS). 05
- c) Apply DFS method to find the Topological sequence for the graph shown below.

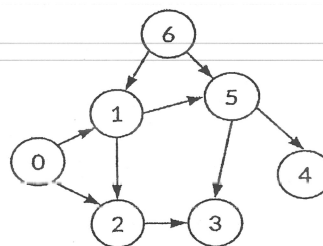


Fig 2(c)

OR

- 3 a) Design an algorithm for performing Sequential search and compute best case, worst case and average case efficiency. 07
- b) Explain Decrease and Conquer and its variations with neat diagram. 07
- c) Design an algorithm for performing Insertion sort. Analyze its time efficiency. Apply the same to sort following set of numbers 4,9,0,-1,6,9,8,2,3,12. 06

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

UNIT - III

- 4 a) Illustrate the tracing of quick sort algorithm for the following set of numbers 1,1,9,9,5,5,6,6 and derive its time complexity. 07

- b) Apply Strassen's matrix multiplication to multiply following matrices. Discuss how this method is better than direct matrix multiplication method. 07

$$\begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 5 \\ 1 & 6 \end{bmatrix}$$

- c) Construct the heap for the list 1,8,6,5,3,7,4 by using bottom up algorithm (Show all the steps). 06

UNIT - IV

- 5 a) Obtain solution for a knapsack problem using dynamic method $M=4$, $n=3$ (w_1, w_2, w_3) = (1, 2, 2) and (p_1, p_2, p_3) = (18, 16, 6). Analyze its efficiency. 08

- b) Apply Floyd's algorithm to the cost matrix given below. Show all necessary steps. Analyze its time efficiency. 07

0	∞	3	∞
2	0	∞	∞
∞	7	0	1
6	∞	∞	0

- c) Define Transitive closure of a directed graph. Write Warshall's algorithm to find transitive closure. 05

OR

- 6 a) Define minimum spanning tree. Explain Prim's algorithm to find the minimum cost spanning tree to the graph shown below. Analyze its time efficiency. 09

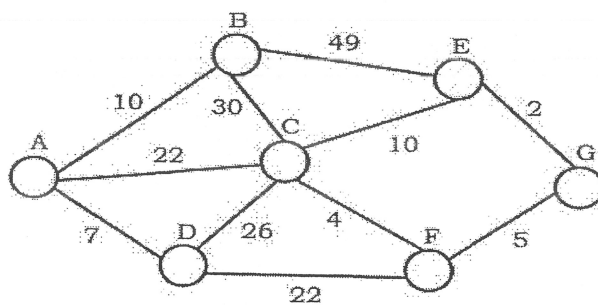


Fig 6(a)

- b) Apply Dijkstra's algorithm to find single source shortest path for the graph given below. Consider Node 6 as source. 09

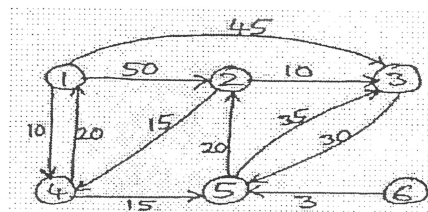


Fig 6(b)

- c) Compare Prim's algorithm and Kruskal's algorithm.

02

UNIT - V

- 7 a) Apply backtracking technique to solve the following instance of the sum of subset problem. $s = \{3, 5, 6, 7\}$ and $d=15$.

07

- b) Apply Branch and bound method to solve knapsack problem and draw state space tree. Given $M=10$

Items	Weight	Value
1	4	40
2	7	42
3	5	25
4	3	12

08

- c) Explain the following

- i) P problem
- ii) NP problem
- iii) NP complete
- iv) NP hard problem

05

B.M.S.C.E. - EVEN SEM 2021-22

